BMAS 1105: MATRICES, DIFFERENTIAL EQUATIONS AND LAPLACE TRANSFORM

Course Objectives: To make the students understand the concepts of matrices, differential equations, and Laplace transform by giving more emphasis to their applications in engineering.

Credits: 4 **Semester II** **L–T–P-J: 3–1–0-0**

Module No.	Contents	Teaching Hours (Approx.)
1	**Matrices**: Introduction, Rank of a matrix by reduction to normal form and echelon form, Consistency and solution of a system of linear equations by matrix method, Complex matrices, Eigen values and Eigen vectors, Applications in engineering. **Ordinary Differential Equations:** Introduction, Solution of exact and reducible to exact differential equations, Solution of n^{th} order linear differential equations with constant coefficients, Euler-Cauchy equations, Applications in engineering.	20
2	**Partial Differential Equations (PDE)**: Introduction, Solution of linear PDE of n^{th} order, Classification of linear PDE of second order. **Laplace Transform**: Introduction, Properties of Laplace transform, Laplace transform of derivatives and integrals, Laplace transform of unit step, Dirac-delta and periodic functions, Inverse Laplace transform and its properties, Convolution theorem, Applications of Laplace transform in solving differential equations, Engineering applications.	20

Focus: This course focuses on employability and skill development aligned with all CO's.

Course Outcomes:

After studying these topics, the student will be able to

- **CO 1:** Know the rank of a matrix and its applications in solving systems of linear equations
- **CO 2:** Find the Eigen values and Eigen vectors of a square matrix
- **CO 3:** Solve linear ordinary and partial differential equations of higher orders
- **CO 4:** Classify the linear partial differential equations as elliptic, parabolic and hyperbolic
- **CO 5:** Apply Laplace transform to Engineering problems using properties
- **CO 6:** Apply Inverse Laplace transform to Engineering problems

Text Books:

- M. Goyal and N. P. Bali, A Text Book of Engineering Mathematics, Laxmi Publication, Delhi, 2014.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi, 2014.

Reference Books:

- W. E. Boyce and R. D. Prima, Elementary Differential Equations, John Wiley & Sons, 2009.
- M. K. Jain, S. R. K. Iyengar and R. K. Jain, Advanced Engineering Mathematics, Narosa Publishing House, Delhi, 2002.